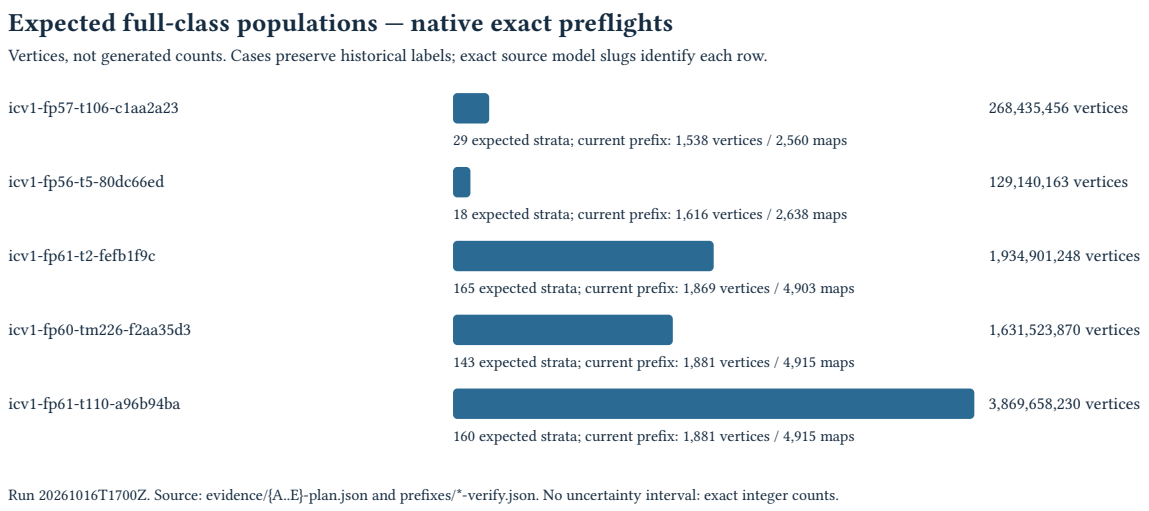
Complete isogeny classes: implementation and coverage

Native enumeration controls and fixed-input preflights, 16 October 2026.

The CLI is implemented and the bounded runs below passed. The full five-case dataset request remains open. A complete vertex class and a complete graph for selected prime degrees are separate claims. Arbitrary isogeny degrees form an infinite family.



Observed correctness evidence

Source model (ICV1)	Vertices	Directed maps	Coverage
icv1-fp7-t2-4910f2d0	8	16	Complete degree-2 class; 4 strata
icv1-fp8-tm1-1ed3feb7	9	18	Complete degree-3 class; 3 strata
icv1-fp7-t4-0c09e92f	10	36	Complete degree-2/3 class; 4 strata

All three vertex sets agree with an independent exhaustive census of every nonsingular model in the corresponding small field. Six native release tests also cover a nontrivial surface class group, checkpoint/resume equivalence, corrupt-map rejection, missing-degree components, BSGS and subgroup/budget checks. Two binary controls also check a killed writer's recovery/resume against an uninterrupted graph and preservation of existing output files. Raw test output: evidence/native-tests.txt. No method timing or speed comparison is admitted by these correctness tests.

Each exact large input was expanded 1,024 times and replayed. Those outputs are partial; unexpanded frontier vertices remain. Replay regenerated every outgoing kernel on expanded vertices and checked every exact rational-map identity and target scaling. It shares native arithmetic with construction. An independent implementation audit of the new generic backend has not been performed.

What the command does

`crypto isogeny class build --curve curve.json --out graph.sqlite --steps all` uses certified root order, exact class populations, complete rational-kernel enumeration and atomic SQLite vertex expansions. The equivalent standalone commands are `isogeny class ...` and `isogeny-class ...`. Automatic completion triggers replay and conductor-population checks. Resume keeps committed expansions. Vertex inspection returns explicit rational maps.

Backend limits: ordinary prime fields 3 < p < 2^61, −10,000,000 <= D_K < −4, prime degrees <=31 different from p. Other cases fail explicitly. The source endomorphism conductor is not inferred from the Frobenius conductor alone.

Full datasets, storage and DLP obligations

The historical degree-3 case has all 129,140,163 vertices, 18 strata and 258,280,326 directed kernel maps in its separately distributed lossless archive. Its compressed bytes and hashes are preserved in evidence/historical-B/. Historical timings were produced outside this repository's measurement protocol.

The five full classes contain 7,833,658,967 vertices and 515 strata. A 24-byte layout for a,b,j alone occupies 188,007,815,208 bytes, excluding edges, indexes, journals and benchmark evidence. This is a fixed-layout baseline, not a measured SQLite size estimate or an information-theoretic lower bound. Compressed or generative encodings may be smaller. The workspace has 32 GiB total. Full remaining class generation needs an appropriate durable execution volume or another validated storage backend; it is not declared impossible.

The solve command verifies one explicit cyclic-subgroup BSGS logarithm. It does not complete the requested graph-wide rho/BSGS/PH and direct/finite-field IC campaign. All-vertex end-to-end timing, total operations, relation yields and min/max/mean remain pending. No sampled result is labeled exhaustive.

Evidence and reproduction: PROTOCOL.md, run-controls.sh, run-prefixes.sh, tools/volcano-cli/README.md, native plans, build/replay JSON and the named storage problem. The existing class-number and ordinary-volcano theorems are reused; no new mathematical theorem or Lean proof is claimed.